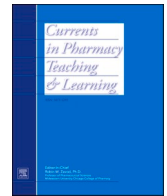




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ChatGPT: Pharmacy students' perceptions, current use trends, ethical awareness, standards of ethics, prospects and recommendations for future use

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ABSTRACT

The launch of ChatGPT by OpenAI reflected a significant turning point in the search and management of diversified information in all fields of knowledge. The concurrent use and ethical awareness of ChatGPT among pharmacy students are examined in detail in this study. During the academic year 2023–2024, a cross-sectional survey was conducted with 413 pharmacy students from different universities in Saudi Arabia. An online questionnaire evaluated their perception and impressions, as well as literacy levels about ChatGPT, together with its proper applications in this professional field of study. Based on the awareness levels of ChatGPT, participants were split into two groups for comparison. It was revealed that a sizable proportion of students (62.55%) were unaware of the concepts and their use. However, students with prior experience using ChatGPT exhibited higher levels of awareness. Nonetheless, on pharmacy-related issues, only 57.7% of participants thought that ChatGPT's responses and information were either somewhat or to some extent appropriate. However, a number of participants (52.3%) had doubts and worries regarding the using ChatGPT. The majority of the participants (78.0%) thought that ChatGPT would have a beneficial effect on pharmacy education regardless of the pending concerns. Only 1.2% and 2.7% of total participants correctly identified the problematic and appropriate applications of ChatGPT, respectively, which indicated an insufficient literacy and ethical awareness about the use of ChatGPT. These results demonstrated a critical need for improved education and training for ChatGPT use, which also warranted more precise recommendations for ethical applications of ChatGPT. It is imperative to fill the educational gaps for a better profession-bound ethical use of ChatGPT. The current ChatGPT use status, including current levels of ethical considerations, guidelines for better ethical use, prospects, and recommendations for enhancing ChatGPT use, are outlined.

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1877-1297/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

A Generative Pre-Trained Transformer (GPT) is a type of large language model that creates text, answers prompt, and converses dynamically.^{1,2} ChatGPT is arguably the most used application of GPT, and its numerous academic and professional uses speak to that. The model's capabilities in natural language understanding and contextually relevant responses have attracted the attention of learners and educators alike, especially in the health disciplines. Incorporating ChatGPT in healthcare education demands the reassessment of conventional evaluation approaches. Generating tailored clinical cases and enhancing topics' comprehensibility, providing fresh possibilities for boosting problem-based learning, and critical thinking.²⁻⁵ The benefits of adopting ChatGPT in academic and scientific writing have been highlighted in several studies.⁶ Such benefits involve strengthening language competence and offering nonstop access to information. Additionally, ChatGPT assists students in research, generating academic papers, and being well-prepared for exams and assignments without drawing the notice of software that detects plagiarism, which goes beyond accepted ethical norms.⁷ A recent study demonstrated ChatGPT's groundbreaking potential in healthcare education. This technology, powered by Artificial Intelligence (AI), potentially transforms the way curricula are tailored in real time. Educators may also utilize ChatGPT to bridge the divide between the rapidly expanding fields of medicine and educational frameworks arena. ChatGPT empowers healthcare students to participate in discussions by offering valuable feedback, stimulating enquiries, and insightful observations in virtual environments.⁸ Thus, the AI-mediated ChatGPT-based debate encouraged pharmacy students to explore multiple points of view, tactics, and medical concepts, thereupon resulting in a more dynamic and participatory concurrent pharmacy education.

ChatGPT has additionally shown excellent performance in plotting a route for several healthcare assessments, including the United States Medical Licensing Examination (USMLE), Saudi Medical Licensing Exam (SMLE), postgraduate ophthalmology examination, medical student parasitology exam, and the American Heart Association (AHA) life support exam.⁹⁻¹³ ChatGPT has also been shown to be more precise in exploring and being trustworthy than the pharmacists in handling factual knowledge questions, implying that it may be helpful for pharmacists.¹⁴ However, it is crucial to consider that ChatGPT's information may not always reflect the most current developments, particularly in rapidly evolving fields like healthcare. Nonetheless, ChatGPT has the potential to improve the learning journey in an array of educational applications, including problem-based learning (PBL), team-based learning (TBL), and case-based learning (CBL).¹⁵ Therefore, using ChatGPT enhance analytical abilities and encourages collaborative team decision-making at its best.⁸ However, it remains imperative to acknowledge the inherent limitations and ethical concerns associated with ChatGPT. These include the potential for biased content generation, transparency and data privacy challenges, susceptibility to manipulation, and the risk of producing inequitable outputs. Copyright and accountability issues often give rise to legal concerns as well.^{2,5,16-19} A phenomenon called "hallucination" may transpire when ChatGPT produces erroneous scientific data that seems plausible to individuals lacking sufficient understanding of the topic. This is where the model generates responses based on patterns in training data without actual comprehension, and this can mislead users who lack sufficient background knowledge on the topic.^{20,21} Sallam et al. (2023) identified risk perceptions, usefulness, convenience of use, attitudes towards technology, and behavioral factors are significant considerations when implementing ChatGPT as a tool in healthcare education.²² Therefore, to guarantee that ChatGPT is used appropriately, many educational institutions worldwide have created standards and guidelines for students to ensure that ChatGPT is utilized responsibly in learning environments. Higher education educators are prepared to incorporate generative artificial intelligence into their curricula to promote professional growth, collaborative learning, and creative teaching methods.²³ In this context, the ethical use of ChatGPT has assumed significant propensity, but currently, there are only a limited number of studies investigating the ethical perspectives of ChatGPT use among a large section of students in healthcare education, while most researchers are now focusing primarily on medical students, the concentration on specific fields of education, such as, allied health sciences, nursing and pharmacy are lacking. Some studies have highlighted the ethical implications of ChatGPT use in medical education, but there is a notable lack of comprehensive, in-depth, specific, and fully addressing the pharmacy students' perspectives on ChatGPT use. Existing literature emphasizes the perceptions of ChatGPT's utility and its effectiveness in educational settings, but it does not investigate in detail the ethical considerations and knowledge gaps that students may have regarding ChatGPT applications in their respective fields of pharmacy education.

This study investigated how pharmacy students in Saudi Arabia perceive the use of ChatGPT through assessments of their beliefs and attitudes toward ChatGPT applications. It addresses the existing literature gaps, ethical expectations, and societal concerns. It also provides valuable insights for educators, policymakers, and institutions that are seeking to implement innovative ChatGPT tools as effectively as needed locally, and perhaps globally. Future outlines and conscientious, professional and rightful, correct use of ChatGPT are also discussed.

Methodology

Research questions and hypothesis

This study explores Saudi Arabian pharmacy students' knowledge, attitude, and ethical awareness of using ChatGPT. It aims to find out how well the students are aware of the fundamental concepts of AI such as natural language processing and machine learning, and whether they can describe the roles and applications of ChatGPT. Moreover, the study investigates how students perceive the ethical aspects of using AI tools both in academic and professional environments. The research maintains that students who can define ChatGPT tend to hold favorable beliefs regarding its usefulness and express high levels of literacy. Furthermore, it hypothesizes that the degree of academia specifically the difference between interns and students in their lower years, can significantly affect students' ethics and knowledge. These hypotheses attempt to delineate areas of AI literacy gaps and serve as the foundation for policy

development and educational interventions in pharmacy education.

Study design, settings, and population

A cross-sectional study design was conducted to investigate the ethical concerns among Saudi Arabian pharmacy students using ChatGPT. This study design was chosen for its effectiveness in obtaining a snapshot of their knowledge and moral perspectives. The Regional Research Ethics Committee in Qassim Province is registered by the National Committee of Bio and Medical Ethics (NCBE), Registration No. H-04-Q-001 on April 1, 2024, and authorized the study ethically. Before administering the survey, participants were oriented to the purpose of the research and given assurances about the privacy of their names. They were made aware that participation in the study was entirely optional. The completion of the questionnaire implied consent. The study targeted pharmacy students in Saudi Arabia during the 2023-2024 academic year, specifically from April 1 to April 20, 2024.

The research aimed to include a broad student population from various regions to analyze the targeted criteria thoroughly. Participants were eligible if they met the criteria: pharmacy students currently enrolled in academic years two through five, or pharmacy interns in Saudi Arabia during the study period, and having at least basic familiarity with ChatGPT. The study excluded non-pharmacy students in Saudi Arabia and pharmacy students in academic years outside of 2 to intern years. To ensure regional and academic diversity, a systematic selection, which formed a selection process to capture the regional and institutional diversity of pharmacy schools across Saudi Arabia's all major regions (e.g., Central, Western, Eastern, Southern, Northern) were represented, thereby ensuring representation from schools with varying curricular structures. The approach aligned with the study's aim of achieving a comprehensive representation of pharmacy students nationwide. The surveys were distributed to students *within* these schools in an equi-distribution pattern (nearly 20% in each region) through online survey platforms and were collaborated with faculty members at each institution to share the survey link within their networks. This method was chosen due to its efficiency in distribution widespread use among the student population.

Sample size and techniques

In Saudi Arabia, over 3,000 pharmacy students graduate annually.²⁴ This study aimed to assess 300 pharmacy students, equating to approximately 10% of all pharmacy graduates. The study aimed to include students from all pharmacy schools across Saudi Arabia to ensure a comprehensive and representative sample. Pharmacy schools were systematically selected to provide regional diversity and include institutions with varying academic structures to reflect a broad perspective of pharmacy students' knowledge and ethics. Given the diverse educational backgrounds of the target population, the authors divided the task of reaching the target group according to the regions of the pharmacy schools. The stratified random sampling method is essential for understanding the pharmacy student population comprehensively, considering inequalities in knowledge and moral issues across various strata. The study's findings are more likely relevant to all pharmacy students in Saudi Arabia due to the assurance of varied regional and academic year coverage. This study's 'academic level' includes pharmacy interns as a separate category to reflect their advanced training stage, distinct from their undergraduate years. This categorization provides insights into the knowledge and ethical awareness of those transitioning from academic learning to professional practice.

Data collection

Google Forms, a website program, was used to create the Likert-scale and multiple-choice responses to the online survey. To guarantee maximum clarity and applicability, an AI investigation specialist checked the questions and provided feedback, which allowed for the refinement and categorization of the questions. Then, to accommodate the linguistic diversity of the participants, the questionnaire was initially drafted in English and translated into Arabic by a bilingual expert to eliminate any ambiguities in wording. Another healthcare professional then verified the translation to ensure linguistic and contextual accuracy. The questionnaire was pre-tested with a pilot group of 10 pharmacy students, and their feedback was used to refine it further. The study questionnaire was distributed through widely utilized social media platforms to maximize participant reach and engagement.

The survey was divided into three parts, the first being the evaluation of pharmacy students' perception of AI, and concentrating on their understanding of fundamental conceptions like raw language processing, machine learning, deep learning, and neural networks; the second, the evaluation of pharmacy students' beliefs and impressions of ChatGPT; and third, the evaluation of literacy regarding the suitable usefulness of ChatGPT.⁴ The questions were primarily developed based on the study's objectives and followed the guidelines and questionnaire types, subjective domains, direction, depths, and concerns, which are referenced appropriately, and available in the AI literature.⁴

Data analyses

The alpha coefficient (Cronbach's alpha) was utilized to estimate the psychometric scale's reliability and the questionnaire items' internal consistency. The results were presented and analyzed. The data were introduced and examined in the layout of explicative method.

At the outset, no prior background information on ChatGPT was provided to participants before the survey to ensure that participants' responses reflected their natural level of awareness and perception. This approach was chosen to capture the baseline knowledge without inserting any external influences from the study investigators. Further, the participants were split into two clusters

based on their answers to a question concerning their understanding of ChatGPT. Group 1 consisted of participants who self-reported that they could define ChatGPT. At the same time, Group 2 included participants who either self-reported being unable to clarify ChatGPT or indicated that they had never heard of it. Participants in Group 2 were included to assess baseline awareness and perceptions of AI tools, such as ChatGPT, even among those who were unfamiliar with its use. This approach provided insights into the knowledge gaps and challenges that may influence future educational interventions and policy development regarding ChatGPT. The respondents' features were corresponded and interpreted in terms of their understanding of the AI, their belief in ChatGPT, and their literacy concerning the suitable usefulness of ChatGPT.

Section 1 consisted of four three-point Likert-scale inquiries. The scoring measures were as follows: -1 = "I have no idea," 0 = "I have heard of it, but I cannot explain it," and 1 = "I know the term and can explain it". The median scores corresponded between the two clusters. Section 2 included nine five-point Likert-scale inquiries. The scoring measures were as; 2 = "Agree", 1 = "Somewhat Agree", 0 = neither "Neither Agree nor Disagree," -1 = "Somewhat Disagree," and, -2 = "Disagree". Participants were also asked to self-rate their knowledge of ChatGPT on a three-point Likert scale, with the options; -1 ('I have no idea'), 0 ('I have heard of it but cannot explain it'), and 1 ('I know the term and can explain it'). Group definitions were based on these self-ratings rather than direct knowledge demonstration. The median scores corresponded between the two clusters. Section 3 contained two multiple-choice inquiries, and the scoring measures were as: choosing the accurate response achieved a score of 1, while not assigning the accurate response resulted in a score of -1. All the data were gathered using Google Forms, and all the investigations were performed employing SPSS Version 26 software (IBM Corporation, New York, NY, USA). Quantitative data are shown as medians (IQR), while categorical data are reported as frequencies (percentages). Kolmogorov–Smirnov (K–S test) was utilized to test for the standard allocation of the constant variables before conducting inferential examinations. Mann-Whitney U was utilized to analyze the discrepancy in medians of the continuous variables (understanding of AI, the belief in ChatGPT, and literacy regarding the suitable usefulness of ChatGPT) between the targeted clusters. Pearson's Chi-square or Fisher's exact test investigated the relationship between categorical variables.

Results

Respondents' survey

The Cronbach's alpha values were 0.672, 0.814, and 0.742 for knowledge of AI, belief in ChatGPT, and literacy concerning the suitable usefulness of ChatGPT, respectively. A total count of 413 students, spread over five years from year 2 to interns, took part in the survey, transcending the originally targeted sampling size of 300. Among the participants, females were prevalent, accounting for 276 students (66.8%), while males accounted for 137 students (33.2%). Concerning their experiences with ChatGPT, 92 students (22.3%) reported using it, while 321 participants (77.7%) declared having never used the program. When questioned if they had heard of ChatGPT, 157 participants (38.0%) expressed they could define it, while 256 participants (62.0%) noted they had only heard of it, though they could not deliver a definition. For students divided into Group 1, which included knowledgeable participants, whereas the other group consisted of those students who were not knowledgeable. The narrative overviews of all the groups are presented in tabular format (Table 1).

Concerning students' knowledge, their acquaintance with public phrases exceeded that of more detailed terms. Particularly, 69.0% of the participants were knowledgeable about the phrase "machine learning," while only 32.0% could convey it. Likewise, 62.4% of the participants were knowledgeable about the phrase "deep learning," whereas just 29.5% could convey it. Concerning the phrases "neural networks" and "natural language processing," 44.5% and 35.6% of the participants were knowledgeable about these phrases, respectively, while only 17.9% and 9.0% could convey them (Table 2).

Table 1
Association between demographic characteristics and awareness of AI.

Awareness (n = 413)				
Variables	Group 1 n (%)	Group 2 n (%)	Total n (%)	p-value
<i>Gender</i>				0.507
Male	49 (35.8)	88 (64.2)	137 (100.0)	
Female	108 (39.1)	168 (60.9)	276 (100.0)	
<i>Academic levels</i>				0.216
2 nd Year	39 (41.5)	55 (58.5)	94 (100.0)	
3 rd Year	36 (43.4)	47 (56.6)	83 (100.0)	
4 th Year	39 (33.6)	77 (66.4)	116 (100.0)	
5 th Year	27 (31.0)	60 (69.0)	87 (100.0)	
Internship	16 (48.5)	17 (51.5)	33 (100.0)	
<i>Experience with ChatGPT use</i>				<0.001 ^a
Yes	92 (100.0)	0 (0.0)	92 (100.0)	
No	65 (20.2)	256 (79.8)	321 (100.0)	

^a Pearson's Chi-square test

Table 2
Analysis of awareness scores among the groups.

Group Item	Group 1 (n = 157)		Group 2 (n = 256)		p-value
	n (%)	Median score [min, max]	n (%)	Median score [min, max]	
Natural language processing	108 (68.8)	0 (-1,1)	39 (15.3)	-1 (-1,1)	<0.001 ^a
Machine learning	148 (94.4)	1 (-1,1)	137 (53.5)	0 (-1,1)	<0.001 ^a
Deep learning	152 (96.8)	1 (-1,1)	106 (41.4)	-1 (-1,1)	<0.001 ^a
Neural networks	121 (77.0)	0 (-1,1)	63 (24.6)	-1 (-1,1)	<0.001 ^a
Total score of section (1)		1 (0,4)		-2 (-4,3)	<0.001 ^a
The number of section (1) scores was ≥ 0 (cutoff value)	157 (100%)		6 (2.4)		<0.001 ^b

^a Mann - Whitney U test; ^b Pearson's Chi-square test.

Perceptions of ChatGPT output

The study collected the initial impressions of participants regarding ChatGPT. When participants were asked about ChatGPT's reliability, 70.5% responded with "reliable" or "somewhat reliable." Likewise, when asked if ChatGPT helps them collect data in a better manner and efficiently, 84.5% of participants replied with "agree" or "somewhat agree." Further, in terms of comprehending their study conclusions, 75.6% of participants revealed that they "agree" or "somewhat agree" and that employed that ChatGPT improved their understanding. Moreover, 78.4% of the participants described that ChatGPT enhanced their educational efficiency, while 78.6% noted that it improved their understanding and acquaintance. However, regarding worries or distrust about utilizing ChatGPT, 52.3% of participants held reservations. Subsequently, the effect of ChatGPT on instruction and career was investigated. When questioned if they thought ChatGPT would thoroughly affect instructions, 78.0% of participants approved or somewhat approved, while 15.0% stayed unbiased. Concerning the sufficiency of answers and data delivered by ChatGPT for collecting data on pharmacy-related topics, only 57.7% revealed that they were acceptable or somewhat acceptable. Likewise, when participants were asked if they thought ChatGPT would help pharmacists enhance their career efficiency, 70.2% answered that it would be beneficial or somewhat beneficial, with 21.1% opting for neutrality (Table 3).

Perceptions of the appropriate use of ChatGPT

To review the participants' literacy concerning the appropriate usefulness of ChatGPT, two kinds of questions were put as multiple-choice inquiries. The first inquiry evaluated participants' knowledge of OpenAI's posted ChatGPT use guidelines. Second, the participants were asked to determine all improper goals in utilizing ChatGPT. Exclusively, five out of 413 participants (1.2%) could recognize all improper goals. The item with the least number of accurate answers, at 23.0%, was "Extraction of big data from websites". The item with the second least number of accurate answers was "Seek medical information" at 39.5%. Relatively more correct answers were given for "Use of ChatGPT in solving complex mathematical formulas through computations", and "Use of ChatGPT to search personal information" at 60.8%, and 78.5%, respectively. However, the item with a moderate number of correct answers was "Students were independent/unsupervised using ChatGPT for self-study," at 49.9% (Table 4).

The students were requested to answer a second inquiry to ensure participants' knowledge of broad precautions when utilizing ChatGPT. Exclusively, 11 of 413 participants (2.7%) could identify all the proper items. Of the three accurate answers, 91.0% of the participants were capable of choosing "At times inaccurate information is generated by ChatGPT", 73.1% were capable of choosing "ChatGPT capacity is limited to provide medical, legal, and professional advice," and 67.8% were capable of choosing "Information provided to ChatGPT can be used for data training". In the second part of the analysis, participants were split into two groups based on their answers concerning their understanding of ChatGPT. Their features were corresponded and investigated regarding their

Table 3
Analysis of impression scores between the groups.

Group Item	Group 1 (n = 157)	Group 2 (n = 256)	p-value
	Median score [min, max]	Median score [min, max]	
Q1. Is the information obtained from ChatGPT trustworthy?	1 (-2,2)	1 (-2,2)	0.057 ^a
Q2. Will the ChatGPT program enhance the effectiveness of gathered information?	2 (-2,2)	1 (-2,2)	0.743 ^a
Q3. Will ChatGPT add to the comprehension of search outcomes?	1 (-2,2)	1 (-2,2)	0.324 ^a
Q4. Will ChatGPT use advanced learning?	1 (-2,2)	2 (-2,2)	0.884 ^a
Q5. Will ChatGPT use add to understanding of the topic under search?	1 (-2,2)	2 (-2,2)	0.435 ^a
Q6. Are there any concerns or queries about using ChatGPT?	1 (-2,2)	0 (-2,2)	0.040 ^a
Q7. Will ChatGPT impact positively on education?	2 (-2,2)	1 (-2,2)	0.031 ^b
Q8. Is the information provided by ChatGPT appropriate for pharmacy-related topics?	1 (-2,2)	1 (-2,2)	0.555 ^a
Q9. Will ChatGPT contribute to pharmacists in improving their professional outcomes?	1 (-2,2)	1 (-2,2)	0.710 ^a
Total score	9 (-12,18)	9 (-18,18)	0.313 ^a
No cutoff values were detected in section (2)*			

^a Mann-Whitney U test; NS=not significant; * Since no cutoff values were detected in this section, further data analysis [n (%)] was deemed unnecessary.

Table 4

Analysis of literacy scores between the two groups.

Group	Group 1 (n = 157)		Group 2 (n = 256)		p-value
Item	Correct answer, n (%)	Median (IQR)	Correct answer n (%)	Median (IQR)	
Assessing literacy regarding the appropriate use of ChatGPT					
Q1. Identify all the improper uses of ChatGPT (True or False)					
A. Extraction of big data from websites	27 (17.2)	-1 (-1,1)	68 (26.6)	-1 (-1,1)	0.028 ^{a_b}
B. Seek medical information	65 (41.4)	-1 (-1,1)	98 (38.3)	-1 (-1,1)	0.529 ^{a_b}
C. Use of ChatGPT for solving complex mathematical formulas through computations	106 (67.5)	1 (-1,1)	145 (56.6)	-1 (-1,1)	0.028 ^{a_b}
D. Use of ChatGPT to search personal information	132 (84.1)	1 (-1,1)	192 (75.0)	1 (-1,1)	0.030 ^{a_b}
E. Students were independent/unsupervised using ChatGPT for self-study	75 (47.8)	-1 (-1,1)	131 (51.2)	1 (-1,1)	0.503 ^{a_b}
Score Q1		1 (-3,3)		-1 (-5,5)	0.370 ^a
Q2. Please select all the correct statements regarding ChatGPT (True or False)					
A. At times inaccurate information is generated by the ChatGPT	146 (93.0)	1 (-1,1)	230 (89.8)	1 (-1,1)	0.277 ^{a_b}
B. Information provided to ChatGPT can be used for data training	109 (69.4)	1 (-1,1)	171 (66.8)	1 (-1,1)	0.579 ^{a_b}
C. Copyright is not an issue when using ChatGPT	110 (70.1)	1 (-1,1)	177 (69.1)	1 (-1,1)	0.843 ^{a_b}
D. The information generated by ChatGPT and other sources can be distinguished	37 (23.6)	-1 (-1,1)	75 (29.3)	-1 (-1,1)	0.204 ^{a_b}
E. ChatGPT provides the most up-to-date information	56 (35.7)	-1 (-1,1)	97 (37.9)	-1 (-1,1)	0.650 ^{a_b}
F. For the same question repeated many times, ChatGPT will generate the same answer	92 (58.6)	1 (-1,1)	140 (54.7)	1 (-1,1)	0.437 ^{a_b}
G. ChatGPT's capacity is limited to providing medical, legal, and professional advice	115 (73.2)	1 (-1,1)	187 (73.0)	1 (-1,1)	0.964 ^{a_b}
Score Q2		1 (-5,7)		1 (-5,7)	0.983 ^a
Total score Q1 + Q2		2 (-6,10)		2 (-8,10)	0.591 ^a

^a Mann - Whitney U test; ^b Pearson's Chi-square test; NS=not significant.

knowledge of AI, beliefs of ChatGPT, and literacy regarding the appropriate use of ChatGPT. Tables 2, 3, and 4 outline the second part of the study. The study revealed that a larger portion of pharmacy students (62.55%) lack awareness about AI concepts, which indicated a potential area for educational improvement.

Discussion

Current status of ChatGPT use

According to a previous study, ChatGPT has shown great promise as a reliable educational tool. It achieved an accuracy rate of 79% in answering 264 clinical pharmacy-related multiple-choice questions. The study also found that ChatGPT's answers and justifications had a 95% concordance rate, with high reproducibility within and between days.¹⁴

Although the benefits of integrating ChatGPT into educational frameworks are recognized, our study indicates low literacy among pharmacy students surveyed. This is observed from a low percentage of correct responses to all inappropriate (1.2%) and appropriate uses (2.7%). This reflects a significant gap in understanding ethical issues and the proper usage of ChatGPT. Furthermore, the analysis showed no significant difference in the overall ethical literacy scores between the two awareness groups, indicating a general deficiency in the comprehensive ethical understanding of ChatGPT. Additionally, a considerable portion of participants (52.3%) expressed concerns or uncertainties regarding using ChatGPT.

The findings from this study provide critical insights into the knowledge and ethical awareness of ChatGPT usage among pharmacy students in Saudi Arabia. The study reveals that a significant portion of pharmacy students (62.55%) lack awareness about AI concepts, which indicates a potential area for educational improvement. This lack of awareness matters because a solid understanding of ChatGPT is essential for future pharmacists who will increasingly rely on AI tools in clinical decision-making, patient care, and pharmacy management. Therefore, enhancing the application of AI pharmacy education is imperative to ensure that students are adequately prepared to integrate these tools into their professional practice responsibly and ethically, ultimately leading to improved healthcare outcomes.^{25,26}

These findings align with previous research that identified significant gaps in AI education among medical students.²⁵ Interestingly, students with experience using ChatGPT demonstrated a greater familiarity with AI concepts, as illustrated in Table 1. The highest

level of awareness was observed among pharmacy interns (48.5%), which suggest that students gain more knowledge about AI as they progress through their studies and gain practical experience. However, this difference was not statistically significant compared to other academic levels.²⁷

Pharmacy students recognize the benefits of integrating ChatGPT into educational frameworks to improve their ability to interpret complex information, a vital skill in pharmacy education. ChatGPT can enhance academic performance by enhancing students' ability to understand and synthesize information effectively. Most participants (78.0%) believe ChatGPT would positively impact pharmacy education. This perception aligns with the views of healthcare workers in Saudi Arabia, who also consider ChatGPT as beneficial for the future of the healthcare system.²⁸ AI-powered adaptive learning platforms can customize education by providing immediate and accurate feedback.^{27,28} However, our study indicates that only 57.7% of the participants found the information and responses provided by ChatGPT on pharmacy-related topics adequate or somewhat adequate.²⁹ In contrast, 75.1% of healthcare workers in Saudi Arabia reported feeling comfortable incorporating ChatGPT into their practice. This suggests that healthcare workers may have more experience with the practical applications and limitations of AI than pharmacy students.²⁹

Shortcomings of current ChatGPT use

Students needed help accurately describing general AI concepts, such as machine learning and deep learning. This issue may be addressed by integrating ChatGPT into the pharmacy curriculum and providing engaging educational workshops and training on basic pharmaceutical sciences and clinical applications. This approach will make learning more interactive and practical, accompanied by clear guidelines to ensure ethical use. In addition, it will help address the language barrier by offering Arabic language support. It will support the students, instructors, and preceptors. Additionally, the absence of well-defined policies for ChatGPT use impacts its adoption and usefulness. There is a need to map out the acceptable and unacceptable practices in ChatGPT domains. Unfamiliarity is another barrier among the students, which can be cleared by workshops and training.

This study has strengths and limitations to consider when interpreting the findings. This is the first study specifically designed to explore the knowledge and ethical awareness of using ChatGPT among pharmacy students in Saudi Arabia. The study involved participants from 23 pharmacy schools, representing a diverse student population. The rigorous methodology, employing stratified random sampling and validated survey instruments, adds to the study's credibility and accuracy. However, the study relied on self-reported data, which may introduce the possibility of response bias as participants may overestimate or underestimate their knowledge and ethical awareness, which can affect the accuracy of the study's findings.

Standards of ethics

The results of the study suggested the role of institutional guidelines which can play towards the accurate, and proper use of ChatGPT. The ethical constraints of the ChatGPT, including its suitable uses and intrinsic shortcomings, were not well understood by the students. While 2.7% of all the students were able to identify all the proper items, only 1.2% of students could recognize the improper strains. To guarantee informed and responsible ChatGPT use in academic and professional context, several international institutions are currently seeking to put comprehensive guidelines, which is warranted by a lack of proper guidelines with regard to the ethical and legal requirements of the use of ChatGPT, and this needs to be supported by the improvements in literacy levels and objectives for the ChatGPT use with further modification on the existing trends and literacy levels. In the context of academics, a sense of moralistic use of ChatGPT exists all the time. The use of ChatGPT is recommended to be used with responsibility, transparency, inclusion, engagement, cohesion, autonomy, safety, unprejudiced, independence, honesty, holistic and foreseeable consideration of the risks, as well as the acceptable extent of benefits, regards to the due rights, non-infringements, accountability, avoidance of the ethical downside, act and conduct socially considered immoral, suppression and feed of non-factual, wrong, unauthenticated, invalidated, sensitive, and unverified information. It is also recommended for generating, formulating, and promoting positive, unbiased and truthful outcomes. Using the techniques also needs consistency with the desired priorities and emerging trends to support the development positively and honestly through ChatGPT. The issues of authorship are also primary to the ethical conduct of ChatGPT. The clarity of the methodology, acceptance of the net resources, including its limitations in circumventing the information and source, ejection of inaccurate and biased outputs as compared against the credible source(s), violations of rights and privacy of individuals, and groups and organizations as well as institutions, and plagiarism, copyright infringement, and other forms of cheating, including not reporting the use of ChatGPT, and national, racial, and gender stereotyping, the inclusion of false information, and denial of credits, are also other ethical considerations while using the AI techniques and primarily ChatGPT. It is also academically dishonest, not condoned by institutional rules and guidelines, and unfair to other students and peers. Any intentional and unintentional representation of AI-generated content, paraphrased, plagiarized, and reused content, fabricated data, including wishful content and data fabrication, as well as using copyrighted content as the original, is also unethical and a potential legal issue. Moreover, ChatGPT cannot cite the sources, raising potential infringement risks of others' rights, and the rights of the internet-sourced/available proceedings and all information in the protected and public domains that need recognition. In this scenario, certain guidelines and broader tips can be helpful in the ethical use of ChatGPT, which includes adhering to the concurrent institutional guidelines, acknowledging the use of any AI tool, critical self-evaluation, and checking the outputs from credible and authenticated sources, limiting the use of ChatGPT and other AI tools and technique as a source of guidance, inspiration, source of motivation, and a handy learning resource to understand and move forward in the subject of interest.²⁹

Maintaining academic integrity while using programs, especially AI-based ones, is crucial. ChatGPT, which offers added advantages, comes under the purview of ethical considerations of technology and software use. While using ChatGPT, students need to be

open about it and provide rightful credit to the content creator. Radboud University's policy demonstrated that using the text and code (s) generated by ChatGPT in an examination without disclosure constituted plagiarism.³⁰ Promoting a culture of independent study and a strong emphasis on academic integrity are necessary to prevent plagiarism. It might be challenging to strike this balance. Still, it can be fixed by adding more tests and using plagiarism detectors to ensure students' work is original and follows the ethics of working and reporting the results. Therefore, a comprehensive educational strategy is essential to bridge the knowledge and ethical awareness gaps in ChatGPT usage among pharmacy students. The survey results advocated filling these gaps in pharmacy students in Saudi Arabia through institutional participation and guidelines development. Incorporating specific AI-based tools and courses, including ChatGPT, into the pharmacy curriculum will certainly help students gain a better grasp of these technologies, with an emphasis on their applications in pharmacy and other inter-related healthcare domains with the awareness of involvement of ethical implications in their use. In addition, interactive workshops need to be established to provide hands-on experience with ChatGPT through actual case studies that draw attention to the promise and limitations of such technology use. Furthermore, developing a strong ethical foundation inside educational institutions is also essential and will provide the exposure, perception and knowledge of the existence of moral use of technologies, as developing as ChatGPT. By incorporating some of the approaches outlined in Table 5, pharmacy education can better equip its students for the prudent and well-informed use of ChatGPT (and other AI tools) in their future professional practices.

Current use status and developing ethical guidelines

The use of ChatGPT and other generative AI technological tools in academic contexts is still developing. While some schools have forbidden ChatGPT, others still need to have such restrictions, according to their feasibility, in place. On the other hand, several organizations are proactively creating rules and regulations to deal with these technologies' morality and appropriate uses. For instance, instructors at Champlain College and the University of Arizona, US, have restrictions regarding the use of ChatGPT for in-class assignments. Similarly, ChatGPT and other generative AI technologies can only be used for report writing with faculty clearance, according to preliminary rules released by Harvard University, and the University of Texas at Austin, both in the US. Students are expected to acknowledge the use of these technologies.^{31–34} Using best practices for utilizing ChatGPT in higher educational contexts, the University of Pretoria, South Africa, has developed multiple recommendations. There is a more thorough approach. In addition to addressing the ethical issues surrounding such technology, these principles seek to improve teaching and learning through AI tools.^{35,36} Additionally, the University of California system in the US is making efforts to offer guidelines for the appropriate applications of generative AI technologies. The intention is to foster creativity and investigation into these technologies while ensuring their applications don't jeopardize confidential data and transgress the license agreements set out by the tool suppliers.³⁷ Most importantly, concerned colleges are proactively creating regulations and standards to guarantee the correct, ethical, and moral use of generative AI technologies, including ChatGPT, in academic contexts as the field continues to advance. The exercise is expected to fill the voids in knowledge, practices, and ethical guidelines to usher into an era of informed, conscientious, and utmost professional use of ChatGPT.^{38,39}

Recommendations

Institutions are urged to develop clear rules, standards, and best practices for employing ChatGPT and other AI tools and generative assistants with integrity. Pharmacy graduates must gain insight during their academic year's program. By Saudi Arabia's Vision 2030, future pharmacists can be prepared to manage their field's needs rapidly in an ever-changing technological landscape by effectively employing these AI technologies and ChatGPT. The recommendations are intended to assist ChatGPT in reaching its full potential, making it a vital instrument in the education and professional development of Saudi Arabian pharmacy students. In this context, the guidelines development and rules of engaging in practices, applications and day-to-day use of ChatGPT and other AI generative tools by other institutions can serve as the referral point.

Table 5
Strategies to fill ChatGPT awareness and ethical use gaps.

Strategy	Description
Integration in Curricula	Incorporate dedicated AI tools modules into the pharmacy curricula to enhance students' understanding of these technologies, their applications in healthcare and the ethical and legal implications.
Interactive Workshops	Provide hands-on experiences with AI tools, including ChatGPT, through interactive workshops and practical case studies, demonstrating both the potential and limitations of these technologies.
Development of Ethical Framework	Establish a robust ethical framework within educational institutions to guide students on responsible AI and ChatGPT uses, emphasizing the importance of verifying ChatGPT and other AI-tools-generated information as well as addressing privacy and data security issues.
Continuous Assessments	Implement continuous assessment and feedback mechanisms to monitor and improve students' understanding and use of ChatGPT as well as other AI tools.
Faculty Training	Provide training for the faculty to effectively teach and integrate AI tools, including ChatGPT into the curricula, ensuring faculty is well-prepared to guide students in the subject areas.

Future directions for ChatGPT use

All educational institutions can benefit from the development of AI-generative tools and ChatGPT in concurrence with the current and future guidelines, which will maximize the professional ethical use of ChatGPT and related AI assistants. The technology is going to establish itself as a valuable tool for student's academic and professional development. The need for implementing institutional self-generated recommendations and adopting the guidelines of other institutions, organizations, and universities will certainly regulate the clean use of the techniques and technology. Additionally, a comprehensive investigation, as well as more research into the expansion of the technology and deeper debates, must be done to fully comprehend the influence of these AI technologies and ChatGPT on national and international pharmacy education, which certainly will help to obtain a sustainable, ethically, and professionally sound use practices for ChatGPT program. As the pharmacy profession continues to grow, Saudi Arabian pharmacy students must embrace the ethical integration of cutting-edge technologies like ChatGPT to thrive in the country's ambitious Vision 2030 goals and to handle the future healthcare landscape in the pharmacy services and education domains.

Conclusions

The study underlines the need for education on ChatGPT concepts and available AI generative tools, including ChatGPT. ChatGPT's prospects, concerns, shortfalls, benefits, and concurrent use with the ethical requirements in pharmacy education, research, and professional development are outlined. ChatGPT and AI generative tools have broader roles as improvers and add-in tools towards achieving larger applications in pharmaceutical education's improvement and pharmacy professionals' outreach status to practice pharmacy for the professional and interested individuals is discussed. The study also underlines the need for improved and inclusive education on ChatGPT concepts in pharmacy curricula as well as ethical practice guidelines development in the pharmacy domains, which circumvents the other AI tools available and under development. The study additionally found that students who had utilized ChatGPT had higher levels of awareness in the pharmacy and inter-related health profession domains, showing that practical experiences boost knowledge and its outreach. Nonetheless, ChatGPT can potentially improve educational outcomes and enhance the professional knowledge base and experience for future work. Notwithstanding, several students also voiced reservations about its proper and ethical implementation, highlighting the need for explicit standards and rigorous ChatGPT and AI tools training with the professional and moral requirements of clear, focused and direct use. As a result, bridging the educational gaps in knowledge, perception, use and ethical tone is critical to students. The information and skills they need to utilize ChatGPT and other generative AI tools ethically to implement them effectively in their lives as pharmacy professionals are highly warranted.

The study acknowledges some of the limitations that could affect how the findings are interpreted. First, because of varying degrees of exposure and comprehension, it is challenging to compare the perceptions of users (Group 1) and non-users (Group 2) regarding ChatGPT. The majority of respondents represented in Group 2, might give answers that are more predicated on conjecture than well-informed judgments, which would reduce the validity of their observations. Furthermore, using self-reported data raises the possibility of biases because it might not fully reflect participants' objective understanding of ChatGPT emphasizing the need for additional study that includes more impartial evaluations.

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Declaration of competing interest

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References

1. OpenAI. OpenAI: Models GPT-3. Accessed on January 14, 2023. Available from: <https://beta.openai.com/docs/models>; 2022.
2. Sallam M. *The Utility of ChatGPT as an Example of Large Language Models in Healthcare Education, Research and Practice: Systematic Review on the Future Perspectives and Potential Limitations*. Published online on February 21, 2023. <https://doi.org/10.1101/2023.02.19.23286155>.
3. Ray PP. ChatGPT: A comprehensive review of background, applications, key challenges, bias, ethics, limitations and future scope. *Internet Things Cyber-Phys Syst.* 2023;3:121–154. <https://doi.org/10.1016/j.iotcps.2023.04.003>.
4. Iwasawa M, Kobayashi M, Otori K. Knowledge and attitudes of pharmacy students towards artificial intelligence and ChatGPT. *Pharm Educ.* 2023;23(1):665–675. <https://doi.org/10.46542/pe.2023.231.665675>.

5. Sallam M, Salim NA, Al-Tammemi AB, et al. ChatGPT output regarding compulsory vaccination and COVID-19 vaccine conspiracy: a descriptive study at the outset of a paradigm shift in online search for information. *Cureus*. 2023;15(2), e35029. <https://doi.org/10.7759/cureus.35029>.
6. AlZaabi A, ALAmri A, Albalushi H, Aljabri R, AalAbdulsalam A.. *ChatGPT Applications in Academic Research: A Review of Benefits, Concerns, and Recommendations*. Published online on August 18. 2023. <https://doi.org/10.1101/2023.08.17.553688>.
7. Hasanein AM, Sobaih AEE. Drivers and consequences of ChatGPT use in higher education: key stakeholder perspectives. *Eur J Investig Health Psychol Educ*. 2023; 13(11):2599–2614. <https://doi.org/10.3390/ejihpe13110181>.
8. Ahmed Y. Utilization of ChatGPT in Medical Education: Applications and Implications for Curriculum Enhancement. *Acta Inform Medica AIM J Soc Med Inform Bosnia Herzeg Cas Društva Za Med Inform BiH*. 2023;31(4):300–305. <https://doi.org/10.5455/aim.2023.31.300-305>.
9. Kung TH, Cheatham M, Medenilla A, et al. Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. *PLOS Digit Health*. 2023;2(2), e0000198. <https://doi.org/10.1371/journal.pdig.0000198>.
10. Aljindan FK, Al Qurashi AA, Albalawi IAS, et al. ChatGPT conquers the Saudi medical licensing exam: exploring the accuracy of artificial intelligence in medical knowledge assessment and implications for modern medical education. *Cureus*. 2023;15(9), e45043. <https://doi.org/10.7759/cureus.45043>.
11. Antaki F, Touma S, Milad D, El-Khoury J, Duval R. Evaluating the performance of ChatGPT in ophthalmology: an analysis of its successes and shortcomings. *Ophthalmol Sci*. 2023;3(4), 100324. <https://doi.org/10.1016/j.xops.2023.100324>.
12. Huh S. Are ChatGPT's knowledge and interpretation ability comparable to those of medical students in Korea for taking a parasitology examination? a descriptive study. *J Educ Eval Health Prof*. 2023;20:1. <https://doi.org/10.3352/jeep.2023.20.1>.
13. Fijačko N, Gosak L, Stiglic G, Picard CT, John Douma M. Can ChatGPT pass the life support exams without entering the American Heart Association course? *Resuscitation*. 2023;185, 109732. <https://doi.org/10.1016/j.resuscitation.2023.109732>.
14. van Nuland M, Erdogan A, Açar C, et al. Performance of ChatGPT on factual knowledge questions regarding clinical pharmacy. *J Clin Pharmacol*. 2024. <https://doi.org/10.1002/jcph.2443>.
15. Wu Y, Zheng Y, Feng B, Yang Y, Kang K, Zhao A. Embracing ChatGPT for medical education: exploring its impact on doctors and medical students. *JMIR Med Educ*. 2024;10, e52483. <https://doi.org/10.2196/52483>.
16. Marchandot B, Matsushita K, Carmona A, Trimaille A, Morel O. ChatGPT: the next frontier in academic writing for cardiologists or a Pandora's box of ethical dilemmas. *Eur Heart J Open*. 2023;3(2), oead007. <https://doi.org/10.1093/ehjopen/oead007>.
17. Liebrezn S, Schleifer R, Buadze A, Bhugra D, Smith A. Generating scholarly content with ChatGPT: ethical challenges for medical publishing. *Lancet Digit Health*. 2023;5(3):e105–e106. [https://doi.org/10.1016/S2589-7500\(23\)00019-5](https://doi.org/10.1016/S2589-7500(23)00019-5).
18. Stokel-Walker C, Van Noorden R. What ChatGPT and generative AI mean for science. *Nature*. 2023;614(7947):214–216. <https://doi.org/10.1038/d41586-023-00340-6>.
19. Lin Z. *Why and How to Embrace AI Such as ChatGPT in Your Academic Life*. Published online February 5. 2023. <https://doi.org/10.31234/osf.io/sdx3j>.
20. Deng J, Lin Y. The benefits and challenges of ChatGPT: an overview. *Front Comput Intell Syst*. 2023;2(2):81–83. <https://doi.org/10.54097/fcis.v2i2.4465>.
21. Shen Y, Heacock L, Elias J, et al. ChatGPT and other large language models are double-edged swords. *Radiology*. 2023;307(2), e230163. <https://doi.org/10.1148/radiol.230163>.
22. Sallam M, Salim NA, Barakat M, et al. Assessing health students' attitudes and usage of ChatGPT in Jordan: validation study. *JMIR Med Educ*. 2023;9, e48254. <https://doi.org/10.2196/48254>.
23. Alammari A. Evaluating generative AI integration in Saudi Arabian education: a mixed-methods study. *PeerJ Comput Sci*. 2024;10, e1879. <https://doi.org/10.7717/peerj-cs.1879>.
24. Ministry of Health. Saudi Arabia. Statistical Yearbook 2022. Accessed February 20, 2024 <https://www.moh.gov.sa/en/ministry/statistics/book/documents/statistical-yearbook-2022.pdf>; 2022.
25. Onesio-Ozigagun O, Ololade YJ, Eyo-Udo NL. Damilola oluwaseun ogundipe. Revolutionizing education through AI: a comprehensive review of enhancing learning experiences. *Int J Appl Res Soc Sci*. 2024;6(4):589–607. <https://doi.org/10.51594/ijarss.v6i4.1011>.
26. Mehta N, Harish V, Bilimoria K, et al. *Knowledge of and Attitudes on Artificial Intelligence in Healthcare: A Provincial Survey Study of Medical Students*. 2021. <https://doi.org/10.1101/2021.01.14.21249830>.
27. Temsah MH, Aljamaan F, Malki KH, et al. ChatGPT and the future of digital health: a study on healthcare workers' perceptions and expectations. *Healthc Basel Switz*. 2023;11(13):1812. <https://doi.org/10.3390/healthcare11131812>.
28. Ambele R, Kaijage S, Dida M, Trojer L, Kyando N. *A Review of the Development Trend of Personalized Learning Technologies and its Applications*. 2022.
29. Stahl BC, Eke D. The ethics of ChatGPT – Exploring the ethical issues of an emerging technology. *Int J Inf Manag*. 2024;74, 102700. <https://doi.org/10.1016/j.ijinfomgt.2023.102700>.
30. Shaw D, Morfeld P, Erren T. The (mis)use of ChatGPT in science and education: Turing, Djerassi, “athletics” & ethics. *EMBO Rep*. 2023;24(7), e57501. <https://doi.org/10.15252/embr.202357501>.
31. LibGuides: Student Guide to ChatGPT: Is Using ChatGPT Considered Cheating?. <https://libguides.library.arizona.edu/students-chatgpt/integrity>. Accessed on February 16, 2024.
32. Blumberg, J. Communicating your ChatGPT/AI Policies - The Center for Learning and Teaching. The Center for Learning and Teaching. <https://clt.champlain.edu/kb/communicating-your-chatgpt-ai-policies/>. Accessed on February 16, 2024.
33. Center for Teaching & Learning. *ChatGPT and Generative AI Tools: Sample Syllabus Policy Statements*. UT Drupal Kit Version 3. Center for Teaching & Learning; 2024. <https://ctl.utexas.edu/ChatGPT-and-generative-ai-tools-sample-syllabus-policy-statements>. Accessed on February 16, 2024.
34. Office of the Provost. *Guidelines for Using ChatGPT and Other Generative AI tools at Harvard*. Office of the Provost; 2024. <https://provost.harvard.edu/guidelines-using-ChaGPT-and-other-generative-AI-tools-harvard>. Accessed on February 16, 2024.
35. Leveraging Generative Artificial Intelligence for Learning. www.up.ac.za/media/shared/391/pdfs/up-student-guide_leveraging-generative-artificial-intelligence-for-learning.zp242396.pdf; 2024. Accessed June 9, 2024.
36. University of Pretoria. <https://www.up.ac.za/media/shared/391/pdfs/up-guide-for-chatgtp-for-teaching-and-learning.zp233629.pdf>. Accessed on February 16, 2024.
37. Office of Ethics. *Appropriate Use of Generative AI tools*. Office of Ethics; 2024. <https://ethics.berkeley.edu/privacy/appropriate-use-generative-AI-tools>. Accessed on February 16, 2024.
38. Farhud DD, Zokaei S. Ethical issues of artificial intelligence in medicine and healthcare. *Iran J Public Health*. 2021. <https://doi.org/10.18502/jpg.v50i11.7600>.
39. Alam F, Lim MA, Zulkipli IN. Integrating AI in medical education: embracing ethical usage and critical understanding. *Front Med*. 2023;10, 1279707. <https://doi.org/10.3389/fmed.2023.1279707>.